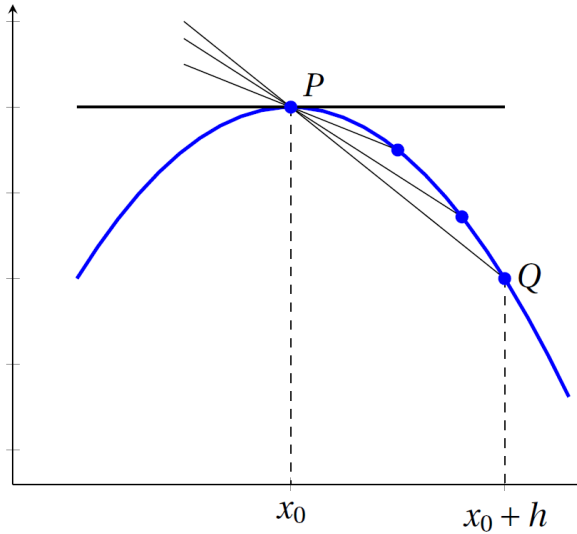


2.1 Tangent Lines and Their Slopes

Problem: Find a straight line L that is tangent to a curve C at a point P .

“For simplicity, restrict ourselves to curves which are graphs of functions.”

How do we define the tangent line to a curve?



The slope of the line PQ is

$$\frac{f(x_0 + h) - f(x_0)}{h}.$$

Definition 2.1.1. Suppose f is cts at $x = x_0$ and

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = m$$

If the limit exists, then the line with equation

$$y = m(x - x_0) + f(x_0)$$

is called **the tangent line** to the graph of $y = f(x)$ at $P = (x_0, f(x_0))$. If the limit does not exist and $m = \infty$ or $m = -\infty$ then the tangent line is the vertical line $x = x_0$. If the limit does not exist and is not $\pm\infty$ then there is no tangent line at P .

Example 2.1.1. *Find an equation of the tangent line to the curve $y = x^2$ at $(1, 1)$.*

Example 2.1.2. *Find an equation of the tangent line to the curve $y = x^{1/3} = \sqrt[3]{x}$ at the origin.*

Example 2.1.3. Does $f(x) = x^{2/3}$ have a tangent line at $(0,0)$?

Example 2.1.4. Does $f(x) = |x|$ have a tangent line at $(0,0)$?

2.2 Derivative

Definition 2.2.1. The *derivative* of a function f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

whenever the limit exists. If $f'(x)$ exists, f is called **differentiable** at x .

$f'(x)$ is the slope of the tangent line to the graph of f at $(x, f(x))$.

We will regard f' as a function whose domain is those x at which f is differentiable.

Another way of defining derivative is

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Two limits are equivalent. This can be seen by letting $x = x_0 + h$.

Example 2.2.1. *Show that the derivative of the linear function $f(x) = ax + b$ is $f'(x) = a$. In particular the derivative of a constant function is zero.*

Example 2.2.2. *Use the definition of the derivative to calculate the derivatives of a) $f(x) = x^2$, b) $f(x) = \frac{1}{x}$, c) $f(x) = \sqrt{x}$.*

The previous three formulas are special cases of the following **Power Rule for Derivative**:

$$f(x) = x^r \implies f'(x) = r x^{r-1}$$

whenever x^{r-1} makes sense.

Proof of the Power Rule for positive integers. Let $f(x) = x^n$ and n a positive integer. Then

$$\begin{aligned} f'(x_0) &= \lim_{x \rightarrow x_0} \frac{x^n - x_0^n}{x - x_0} = \lim_{x \rightarrow x_0} \frac{(x - x_0)(x^{n-1} + x^{n-2}x_0 + \cdots + x x_0^{n-1} + x_0^{n-1})}{x - x_0} \\ &= \lim_{x \rightarrow x_0} (x^{n-1} + x^{n-2}x_0 + \cdots + x x_0^{n-1} + x_0^{n-1}) = n x_0^{n-1} \end{aligned}$$

We will prove the general version later.

Example 2.2.3.

$$f(x) = x^{5/3} \implies f'(x) = x^{2/3},$$

for all x . How about $f'(-1/8)$?

$$f(x) = \frac{1}{\sqrt{x}} \implies f'(x) = -\frac{1}{2}x^{-3/2}$$

for $x > 0$.

Example 2.2.4. *Differentiate the absolute value function $f(x) = |x|$ to get*

$$f'(x) = \operatorname{sgn}(x) = \begin{cases} -1, & \text{if } x < 0 \\ 1, & \text{if } x > 0 \end{cases}$$

Note that f is not differentiable at 0.

Example 2.2.5. *How should the function $f(x) = x \operatorname{sgn}(x)$ be defined at $x = 0$ so that it is continuous there? Is it then differentiable there?*

Notations for Derivative

Let $y = f(x)$. We denote the derivative by

$$y' = f'(x) = \frac{dy}{dx} = \frac{d}{dx}f(x).$$

If we want to evaluate the derivative at point x_0

$$y' |_{x=x_0} = f'(x_0) = \frac{dy}{dx} |_{x=x_0} = \frac{d}{dx}f(x) |_{x=x_0} .$$

The notations y' and $f'(x)$ are *Lagrange notations* for the derivative. The notations $\frac{dy}{dx}$ and $\frac{d}{dx}f(x)$ are called *Leibniz notations* for the derivative.

The Leibniz notation is suggested by the definition of the derivative. Let $\Delta y = f(x+h) - f(x)$ be the increment in y and $\Delta x = x+h - x = h$ be the increment in x . Then

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

2.3 Differentiation Rules

Differentiability is stronger than continuity.

Theorem 2.3.1. *If f is differentiable at x then f is cts at x .*

Proof.

$$\lim_{h \rightarrow 0} (f(x+h) - f(x)) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \lim_{h \rightarrow 0} h = f'(x)0 = 0$$

This means

$$0 = \lim_{h \rightarrow 0} f(x+h) - \lim_{h \rightarrow 0} f(x) = \lim_{h \rightarrow 0} f(x+h) - f(x)$$

Hence

$$\lim_{h \rightarrow 0} f(x+h) = f(x)$$

Theorem 2.3.2. *If f and g are differentiable at x then*

$$(f + g)'(x) = f'(x) + g'(x),$$

$$(f - g)'(x) = f'(x) - g'(x),$$

and for any constant c

$$(cf)'(x) = cf'(x).$$

Proof. Let's prove the derivative of sums is sum of derivatives. The others are similar.

$$\begin{aligned}(f + g)'(x) &= \lim_{h \rightarrow 0} \frac{(f + g)(x + h) - (f + g)(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x + h) + g(x + h) - f(x) - g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x + h) - g(x)}{h} = f'(x) + g'(x),\end{aligned}$$

The sum rule extends to any number of functions.

$$(f_1 + \cdots + f_n)'(x) = f_1'(x) + \cdots + f_n'(x).$$

Example 2.3.1. *Take the derivative of*

$$f(x) = 5\sqrt{x} + \frac{3}{x} - 19$$

Theorem 2.3.3. *If f and g are differentiable at x then*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Example 2.3.2. *Find the derivative of $f(x) = (x^2 + x + 1)(2x + \frac{1}{x})$.*

Theorem 2.3.5. *If f and g are differentiable at x and $g(x) \neq 0$ then*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

Example 2.3.3. *Differentiate $y = \frac{x^5}{x^{2/3} + 1}$.*

Example 2.3.5. Find an equation of the tangent line to $y = \frac{2}{3 - 4\sqrt{x}}$ at the point $(1, -2)$.

Example 2.3.6. Find the x -coordinates of points on the curve $y = \frac{x+1}{x+2}$ where the tangent line is parallel to the line $y = 4x$.

Example 2.3.7. *If $f(2) = 2$ and $f'(2) = 3$, calculate*

$$\left. \frac{d}{dx} \left(\frac{x^2}{f(x)} \right) \right|_{x=2}$$

2.4 Chain Rule

The following theorem is known as the chain rule.

Theorem 2.4.1. *If $f(u)$ is differentiable at $u = g(x)$ and $g(x)$ is differentiable at x , then*

$$(f \circ g)'(x) = f'(g(x))g'(x)$$

Proof. The proof in the case $g(x) \neq g(a)$ for x sufficiently close to a .

$$(f \circ g)'(a) = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a} = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} = f'(g(a))g'(a)$$

If there is always an $x \neq a$, x near a such that $g(x) = g(a)$ then the above proof fails due to division by zero. You can find the proof in this case in Wikipedia. $\square$

In Leibniz notation, if $y = f(u)$ where $u = g(x)$ then

$$y = f(g(x)) = (f \circ g)(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

Example 2.4.1. Find the derivative of $y = \sqrt{x^2 + 1}$.

Example 2.4.2. Differentiate $y = (x^3 - 1)^{1000}$.

Example 2.4.3.

$$\frac{d}{dx} |x| = \frac{d}{dx} \sqrt{x^2} = \frac{1}{2\sqrt{x^2}} 2x = \frac{x}{\sqrt{x^2}} = \frac{x}{|x|}, \quad x \neq 0$$

This function is called sign function or signum function.

$$\operatorname{sgn}(x) = \frac{x}{|x|} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}.$$

Example 2.4.4. *Express in terms of f and f' .*

a) $\frac{d}{dx}f(x^2),$

b) $\frac{d}{dx}(f(\pi - 2f(x)))^4.$

Example 2.4.5. *For*

$$f(x) = \left(1 + \sqrt{2x+1}\right)^{-4/3}$$

evaluate $f'(0)$.

Example 2.4.6. *Find an equation of the tangent line to the graph of*

$$y = (1 + x^{2/3})^{3/2}$$

at $x = -1$.

2.5 Derivatives of Trigonometric Functions

The radian measure of an angle is defined to be the length of the arc of a unit circle corresponding to that angle.

$$\text{angle in degrees} = \text{angle in radians} \cdot \frac{180^\circ}{\pi}.$$

In calculus all angles are measured in radians. By an angle of $\pi/3$ we mean $\pi/3$ radians or 60° not $(\pi/3)^\circ \approx 1.04^\circ$.

Theorem 2.5.1. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$

Example 2.5.1. *Show that $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$.*

Example 2.5.2. *Find the limit of*

- $\lim_{x \rightarrow 0} \frac{\sin x}{\sin 2x}$
- $\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x}$
- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$
- $\lim_{x \rightarrow 0} \frac{\tan 2x}{x}$

Theorem 2.5.2. $\sin x$ is differentiable for every x and

$$\frac{d}{dx} \sin x = \cos x$$

Theorem 2.5.3. $\cos x$ is differentiable for every x and

$$\frac{d}{dx} \cos x = -\sin x.$$

Example 2.5.3. Evaluate the derivative of

a) $\sin(\pi x) + \cos(3x),$

b) $x^2 \cos(\sqrt{x}),$

c) $\frac{\cos x}{1 - \sin x}$

The derivatives of the other trigonometric functions

$$\tan x = \frac{\sin x}{\cos x}, \quad \sec x = \frac{1}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \csc x = \frac{1}{\sin x}.$$

Since $\cos$ and $\sin$ are everywhere differentiable, the above functions are differentiable everywhere except where their denominators are zero. The derivatives of these functions can be derived by using quotient and reciprocal rules.

$$\frac{d}{dx} \tan x = \sec^2 x, \quad \frac{d}{dx} \sec x = \sec x \tan x, \quad \frac{d}{dx} \cot x = -\csc^2 x, \quad \frac{d}{dx} \csc x = -\csc x \cot x.$$

Exercises

1. $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$. (Hint: Use Sandwich Theorem)
2. $\lim_{t \rightarrow \infty} \frac{2 - t + \sin t}{t + \cos t}$ (Hint: Divide both sides by t and use the previous exercise)

2.6 Higher Order Derivatives

Derivative of derivative is called **second derivative**. If $y = f(x)$ then

$$y'' = f''(x) = \frac{d}{dx} \frac{d}{dx} y = \frac{d^2}{dx^2} y = \frac{d^2}{dx^2} f(x).$$

Similar notations can be used for third, fourth, etc. derivatives. For n-th derivative, we write

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n}$$

Example 2.6.1. *Calculate all the derivatives of $y = x^3$.*

2.7 Mean Value Theorem

The Mean Value Theorem is the midwife of calculus - not very important or glamorous by itself, but often helping to deliver other theorems that are of major significance.

– E. Purcell and D. Varberg

Suppose you drive in 2 hours from city A to city B which are 200km apart. That means your average speed was 100km/h. Even if you did not travel constant speed, there was at least one instant where your speed was exactly 100km/h. This is called **mean value theorem**.

Theorem 2.7.1 (The Mean-Value Theorem). *Suppose that f is continuous on the interval $[a, b]$ and that it is differentiable on the open interval (a, b) . Then there exists a point c in the open interval (a, b) s.t.*

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

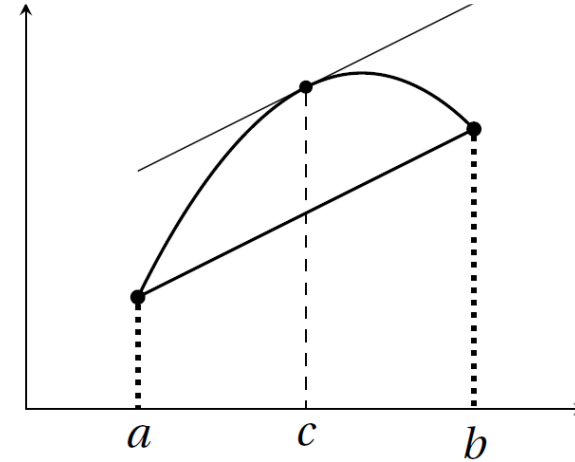


Figure 2.1: Mean Value Theorem says that the slope of the secant line joining two points on the graph of $f(x)$ is equal to the slope of the tangent line at some point $x = c$ between a and b .

Let $f(t)$ denote the distance from city A. Then $f(0) = 0$ and $f(2) = 200$. Mean Value Theorem says there is a time $t = c$ s.t. $f'(c) = 100$.

Example 2.7.1. Let $f(x) = |x|$ on $[-1, 1]$. Show that there is no $c \in [-1, 1]$ satisfying the conclusion of the Mean Value Theorem. Why?

The Mean Value Theorem is an *existence theorem* like Intermediate Value Theorem. In particular

- We don't know how to find c .
- We don't know how many different c can be found satisfying Mean Value Theorem (there is at least one).

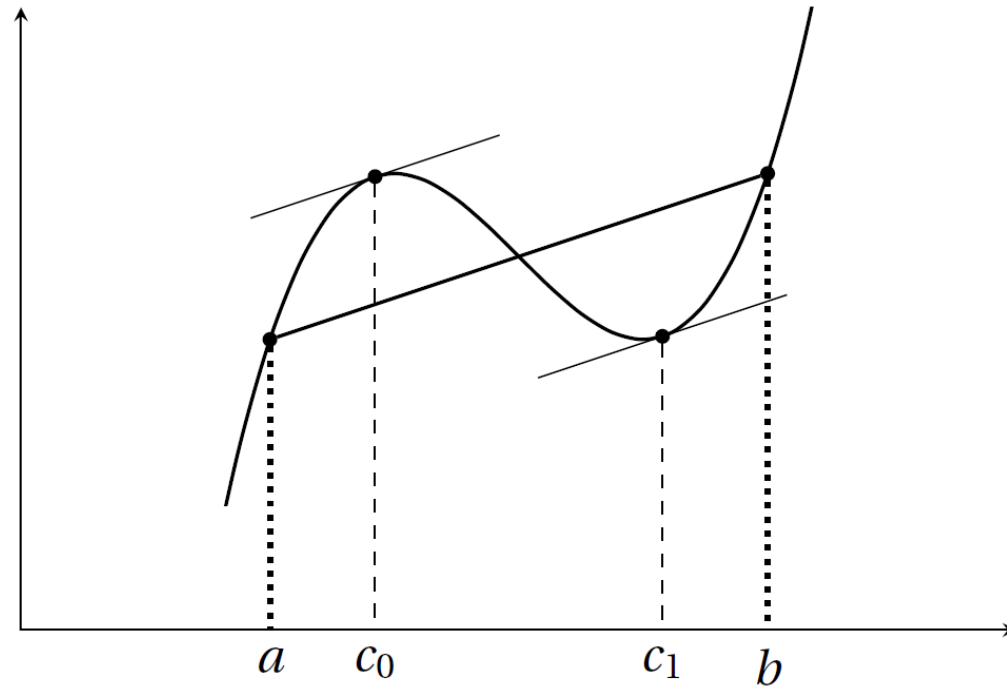


Figure 2.2: There may be more than one c satisfying the conclusion of the Mean Value Theorem.

Example 2.7.2. *Show that $\sin x < x$ for all $x > 0$.*

Example 2.7.3. *Show that $\sqrt{1+x} < 1 + \frac{x}{2}$ for all $x > 0$.*

Example 2.7.4. *Determine all the numbers c which satisfy the conclusions of the Mean Value Theorem for*

$$f(x) = x^3 + 2x^2 - x, \quad x \in [-1, 2]$$

Example 2.7.5. *Suppose f is continuous and differentiable on $[3, 9]$. Suppose $f(3) = -4$, and $f'(x) \leq 10$ for all x . What is the largest value possible for $f(9)$?*

Definition 2.7.1. Suppose f is defined on an interval I . If for all x_1, x_2 in I s.t. $x_2 > x_1$,

If	Then on I , f is
$f(x_2) > f(x_1)$	increasing
$f(x_2) < f(x_1)$	decreasing
$f(x_2) \geq f(x_1)$	non-decreasing
$f(x_2) \leq f(x_1)$	non-increasing

Theorem 2.7.2. Suppose f is differentiable on an open interval I . If for all $x \in I$,

If	Then on I , f is
$f'(x) > 0$	increasing
$f'(x) < 0$	decreasing
$f'(x) \geq 0$	non-decreasing
$f'(x) \leq 0$	non-increasing

Example 2.7.6. *On what intervals is $f(x) = x^3 - 12x + 1$ increasing or decreasing?*

Theorem 2.7.3. *If $f'(x) = 0$ on an interval I then $f(x)$ is constant on I .*

Proof. Choose x_0 in I . Let $C = f(x_0)$. If x is any other point in I then by Mean Value Theorem, $f(x) - f(x_0) = f'(c)(x - x_0) = 0$. $\square$

Challenge Problem. Suppose that $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable functions such that the limits $L = \lim_{x \rightarrow -\infty} f(x)$ and $M = \lim_{x \rightarrow -\infty} g(x)$ exist and $L \leq M$. Is it possible that $f(1) = 4$, and $g(1) = 2$? **Hint:** Consider $h(x) = g(x) - f(x)$ and use MVT.

2.8 Implicit Differentiation

We learned to find the slope of a curve that is the graph of a function. But not all curves are graphs of functions, for example the circle $x^2 + y^2 = 1$.

Curves are graphs of equations in two variables

$$F(x, y) = 0.$$

For the circle $F(x, y) = x^2 + y^2 - 1$.

Example 2.8.1. *Find the slope of the circle $x^2 + y^2 = 25$ at the point $(3, -4)$.*

Example 2.8.2. Find an equation of the tangent line to the curve $x \sin(xy - y^2) = 0$ at $(1, 1)$

Example 2.8.3. Find y'' in terms of x and y if $xy + y^2 = 2x$.

The General Power Rule for Derivative

So far, we proved the following rule

$$\frac{dx^r}{dx} = r x^{r-1}$$

Sample Exam 1

1. Find the following limits if they exist.

a)

$$\lim_{x \rightarrow \infty} \left(\frac{x^2}{x+1} - \frac{x^2}{x-1} \right)$$

b)

$$\lim_{x \rightarrow 0} \frac{x}{|x-1| - |x+1|}$$

c)

$$\lim_{x \rightarrow -\infty} \left(x + \sqrt{x^2 - 4x + 1} \right)$$

3. Find the slope of the tangent line to the curve

$$\tan(xy^2) = \frac{2xy}{\pi}$$

at the point $(-\pi, 1/2)$.

4. Calculate the derivatives.

a)

$$y = \frac{1 + \sqrt{x}}{x^4 + 1}$$

b)

$$y = (\sin(\sqrt{x}) + 1)^3 + 7x \cos x$$

5. Is the function $y = |x - 2|$ differentiable at $x = 2$? Show your work using the limit definition of the derivative.

3.1 Inverse Functions

Definition 3.1.1. f is called **one-to-one** if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ or equivalently

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

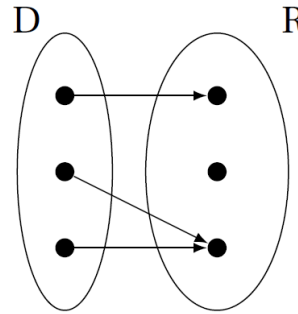


Figure 3.1: A function which is not 1-1.

Horizontal Line Test. Let $f : \mathbb{R} \rightarrow \mathbb{R}$. By definition of a function any vertical line intersects the graph at one point. f is 1-1 if its graph is never intersected by any horizontal line more than once.

Theorem 3.1.1. *Increasing or decreasing functions are 1-1. Thus if $f'(x) > 0$ for all x in an interval I , then f is 1-1 on I . Similarly if $f'(x) < 0$ for all x in I then f is 1-1 on I .*

Definition 3.1.2. *If f is one-to-one then it has an inverse function f^{-1} defined as follows: If x is in the range of f then it is in the domain of f^{-1} and*

$$f^{-1}(x) = y \iff x = f(y).$$

If f is not 1-1 then it is not invertible.

Given $y = f(x)$, to find the inverse function, we solve x in terms of y .

Example 3.1.1. *Show that $f(x) = 2x - 1$ is one-to-one and find its inverse $f^{-1}(x)$.*

Properties of inverse functions

1. The domain of f^{-1} is the range of f .
2. The range of f^{-1} is the domain of f .
3. $f(f^{-1}(x)) = x$ for all x in the domain of f^{-1} .

Proof. If $f^{-1}(x) = y$ then $x = f(y)$ and $f(f^{-1}(x)) = f(y) = x$.

4. $f^{-1}(f(x)) = x$ for all x in the domain of f .
5. $(f^{-1})^{-1}(x) = f(x)$ for all x in the domain of f . (The inverse of inverse of f is f .)

6. The graph of f^{-1} is the reflection of the graph of f in the line $x = y$. (Because if (a, b) is a point on the graph of $y = f(x)$ then (b, a) is a point on the graph of $y = f^{-1}(x)$).

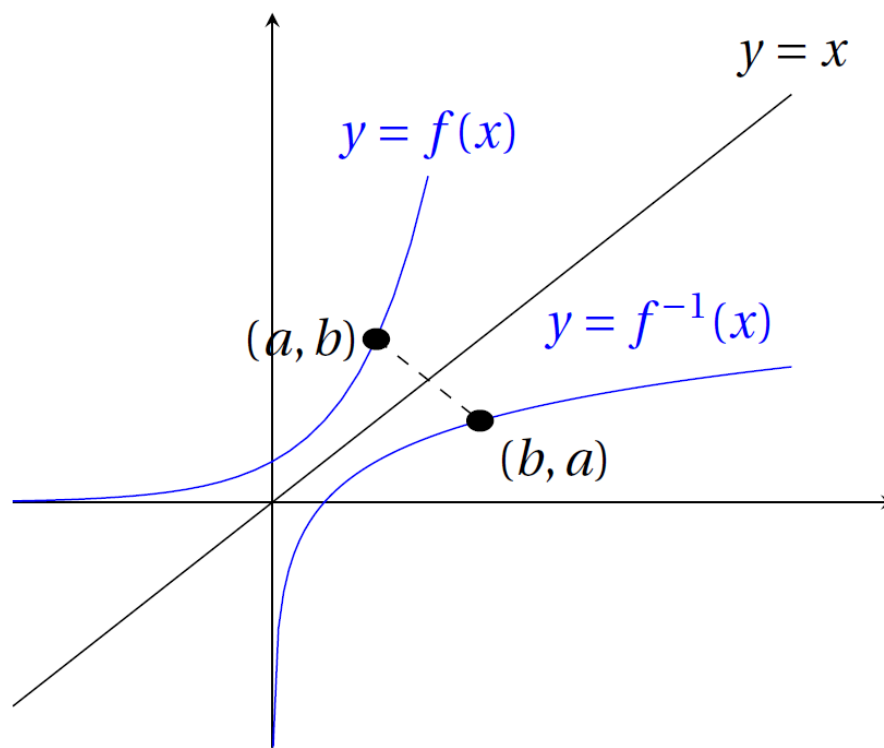


Figure 3.2: The graph of the inverse function is a reflection along $y = x$.

Derivatives of Inverse Functions

Let $y = f^{-1}(x)$. Then $f(y) = x$. Considering y as a function of x and using implicit differentiation,

$$\frac{d}{dx}f(y) = \frac{d}{dx}x \implies f'(y)\frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{f'(y)}$$

In short if $y = f^{-1}(x)$ then

$$(f^{-1})'(x) = \frac{1}{f'(y)} = \frac{1}{f'(f^{-1}(x))}$$

Example 3.1.2. *Show that $f(x) = x^3 + x$ is one-to-one on the whole real line and find $(f^{-1})'(10)$. Hint: $2^3 + 2 = 10$.*

Example 3.1.3. *If $f(x) = 3x + x^3$, show that f has an inverse and find the slope of $y = f^{-1}(x)$ at $x = 0$.*

Theorem 3.3.1. *If $x > 0$ then $\frac{d}{dx} \ln x = \frac{1}{x}$*

Example 3.3.1. *Find the derivatives of*

1. $y = \ln |\cos x|$

2. $y = \ln(x + \sqrt{x^2 + 1})$

Definition 3.3.2. $e = \exp(1) \approx 2.718\dots$

Thus $\ln e = 1$. Hence e is the number for which the area bounded by $y = 1/x$, the x-axis and the lines $x = 1$, $x = e$ is 1.

$$e^x = \exp(\ln(e^x)) = \exp(x \ln e) = \exp(x).$$

Since $\exp$ is actually an exponential function, its inverse must be a logarithm

$$\ln x = \log_e x$$

The derivative of $y = e^x$ is calculated by implicit differentiation:

$$y = e^x \iff x = \ln y \iff 1 = \frac{y'}{y} \iff y' = y = e^x$$

This is a remarkable property:

$$\frac{d}{dx} e^x = e^x, \quad \int e^x dx = e^x + C$$

Example 3.3.2. *Find the derivatives of*

1. $e^{x^2-3x},$

2. $\sqrt{1+e^{2x}}$

Derivative of $y = a^x$.

$$\frac{d}{dx} a^x = \frac{d}{dx} e^{x \ln a} = e^{x \ln a} \ln a = a^x \ln a.$$

Example 3.3.4. *Show that the graph of $f(x) = x^\pi - \pi^x$ has negative slope at $x = \pi$.*

Logarithmic Differentiation

Definition 3.3.4. Let $y = a^x$. Then $\frac{dy}{dx} = a^x \ln a$ which is negative if $0 < a < 1$ and positive if $a > 1$. Thus a^x is 1-1 and has an inverse function. We define its inverse as $\log_a x$.

Derivative of $y = \log_a x$.

$$\frac{d}{dx} \log_a x = \frac{d}{dx} \frac{\ln x}{\ln a} = \frac{1}{x \ln a}.$$

Example 3.3.5. Let $y = x^x$, $x > 0$. Find y' .

Example 3.3.6. Find dy/dt if $y = (\sin t)^{\ln t}$ where $0 < t < \pi$.

Example 3.3.7. If $y = \frac{(x+1)(x+2)(x+3)}{\sqrt{x+4}}$, find y' .

Theorem 3.4.1.

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}. \quad \frac{d \arctan(x)}{dx} = \frac{1}{1+x^2} \quad \frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$$

Example 3.4.4. Find the slope of the curve $\arctan\left(\frac{2x}{y}\right) = \frac{\pi x}{y^2}$ at the point $(1, 2)$.

Example 3.4.6. Show that $\frac{d}{dx} (\arcsin x^2)^{1/2}$